

E1

Consider the circuit in Fig. 1. The circuit parameters, at $T = 25^\circ\text{C}$, are the following:

$V_{CC} = 24\text{ V}$; $R_{B1} = 82\text{ k}\Omega$; $R_{B2} = 120\text{ k}\Omega$; $R_{E1} = 3.9\text{ k}\Omega$; $R_C = 5.6\text{ k}\Omega$; $R_F = 10\text{ k}\Omega$; $R_s = 1\text{ k}\Omega$;
 $C_s = 100\text{ }\mu\text{F}$; $C_E = 1\text{ }\mu\text{F}$; $C_F = 100\text{ }\mu\text{F}$.

Q_1 (nnp BJT): $V_{BE1} = 0.7\text{ V}$; $\beta_{F1} = \beta_{01} = 200$; $r_{01} = +\infty$;

Q_2 (pnp BJT): $V_{BE2} = -0.7\text{ V}$; $\beta_{F2} = \beta_{02} = 150$; $r_{02} = +\infty$;

$V_T = 26\text{ mV}$.

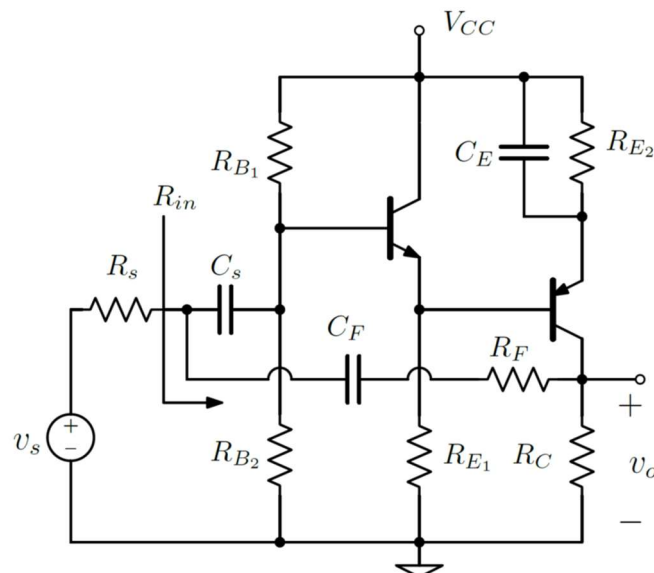


Fig. 1 – A case of two stage amplifier.

Considering all capacitors to be equivalent to *open circuits*, determine:

1. the value of R_{E2} that guarantees the bias level of the output is $V_O = 10.5\text{V}$;
2. the operating points (V_{CE} , I_C) of Q_1 and Q_2 .

Assuming C_s , C_E and C_F to be equivalent to *short circuits* at the frequencies of interest (i.e. at mid-band), determine also:

3. the voltage gain $A_v = v_o/v_s$ of the amplifier;
4. the input resistance indicated in the figure.

Assuming only C_E is effective (while C_s , C_F are practically shorted) determine, finally:

5. the low frequency bandwidth limit f_L of the amplifier.

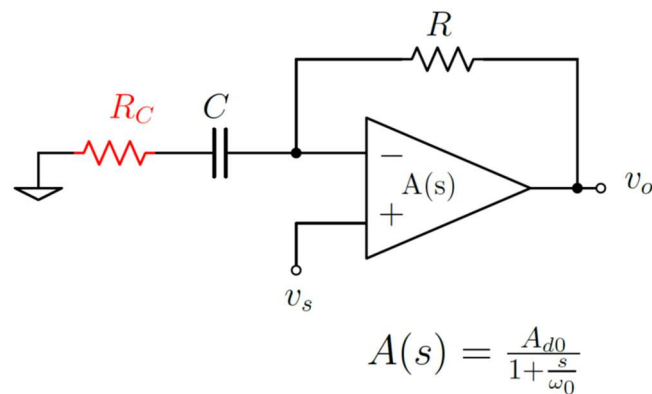


Fig. 2 – OpAmp based amplifier.

Consider the operational amplifier configuration shown in Fig. 2. The circuit parameters are the following:

$R = 10 \text{ k}\Omega$; $C = 10 \text{ nF}$;

OpAmp: $A_{d0} = 10^5 \text{ [V/V]}$; $\omega_0 = 500 \text{ rad/s}$.

Considering initially $R_C = 0 \text{ }\Omega$, determine:

1. the expression of the *ideal* circuit voltage gain $A_v(s) = v_o/v_s$;
2. an estimation of the circuit *phase* margin;
3. a *block diagram* of the amplifier, according to *feedback theory*.

Then, find

4. the maximum value of R_C for which the amplifier maintains *approximately* the *same frequency response* of the original circuit *up to the crossover frequency*;
5. an estimation of the *new phase margin* of the circuit;
6. an alternative, equivalent compensation strategy, sketching the resulting circuit schematic (no calculations required).